Introduction To CNTFET Technology For Reducing Transistor Count And Power Requirement Of ICs

This article introduces a mixed logic design method that uses CNTFET technology combining the basic transmission gate logic, the pass transistor dual-value logic, and base CMOS logic. 14T and 15T configurations have been studied and analysed—14T for 2 to 4 decoder with low transistor count benefit and 15T for high performance in terms of power and delay using 15 transistors. The results are quite encouraging

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The very large scale integration (VLSI) technology has made it possible to fabricate millions of transistors on a single chip. As nanometer process innovations have progressed, making low power circuits has become important in view of tremendous demand for low power chips. This article describes how circuits can be implemented through carbon nanotube field effect transistor (CNTFET), a developing technology that can enhance conventional complementary metal-oxide-semiconductor (CMOS) 32nm chips through better power dissipation and performance.

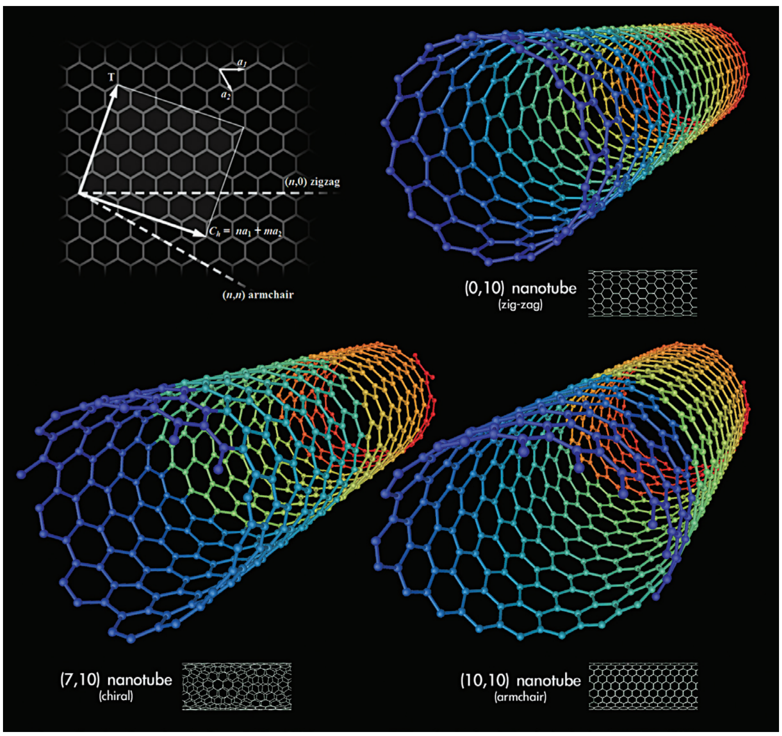

Fig. 1: Types of CNTFET

In most integrated circuit designs the CMOS technique is used at present, beginning with basic digital circuits to a system on chip (SoC). Both N-channel enhancement mode metal and N-channel enhancement mode technology are used in this technique. Majorly, NMOS is used as the pull-down network and PMOS as the pull-up network to implement the circuit. This configuration produces good performance, resistance to noise and device variation. The improvement in parameters like power, performance, cost, and time to advertise have not changed since the origin of the IC business. Truth be told, Moore's Law is tied to advancing these parameters.

In any case, when scaling goes below 20nm, a portion of the circuit parameters cannot be scaled any further, particularly the power supply voltage, which is most important for deciding the power dissipation. Another restriction as procedures move toward 20nm is the lithography process,